

Experiment

Study of LAN Topology Network devices
Address Configuration.

Objective

- 1) To study the LAN Topology used in College Campus.
- 2) To identify networking device & understand their functions.
- 3) To analyze wired & wireless communication method used in the network.
- 4) To measure speed of network for wired & wireless connections.
- 5) To obtain MAC, IP Address & Subnet mask.

Problem Statement-1

Draw & identify LAN Topology of College Campus clearly label all components such as routers, switches & etc.

Soln:-

The College Campus network observed this study consists of -

- * Largest - 3 Core switches in the server room.
- * Managed by PoE + switches.

* Wireless Access Point (AP)

* Ethernet Port in Classroom & Laboratories

Network Topology :-

- (1) Core layer.
- (2) Distribution layer.
- (3) Access layer.

These all are the network hierarchical layers.

Hybrid Hierarchical Topology which combines,

- (i) Tree topology.
- (ii) Extended Star topology.
- (iii) Ring topology (Server room).

Core layer :-

Core layer contains Server Room and Server room have Firewall, Router, 2-13 + Switches Patch, Rack, Server, inverter etc.

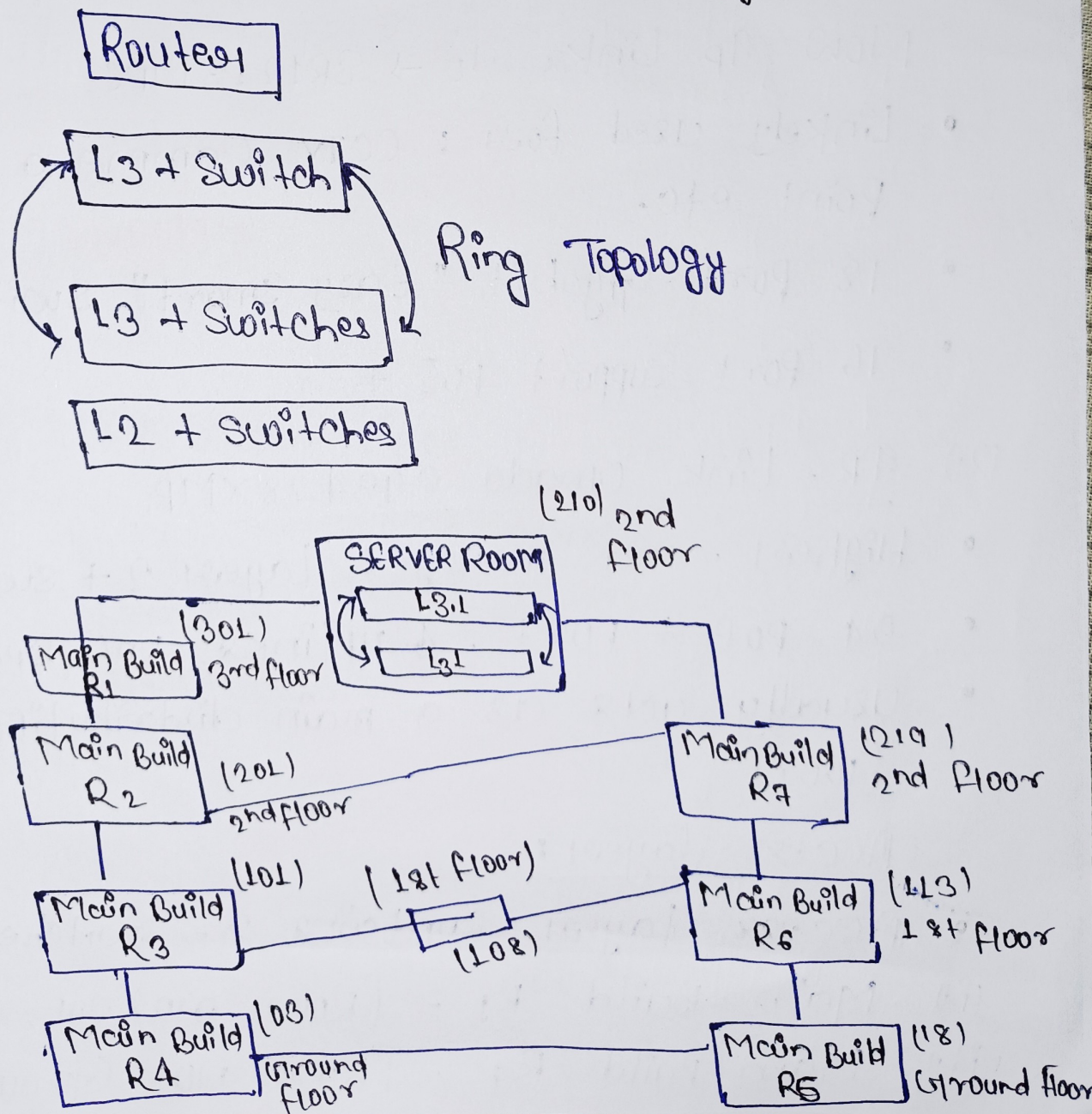
Distribution layer :-

- Distribution layer contains Server Room and
- The 4 Switches are -
 - (i) Main Build → R₇ (Room no. 213, 2nd floor)
 - (ii) Main Build → R₁ (Room no. 301, 3rd floor)

iii) Main Build R5 - [Room no-18, Ground Floor]

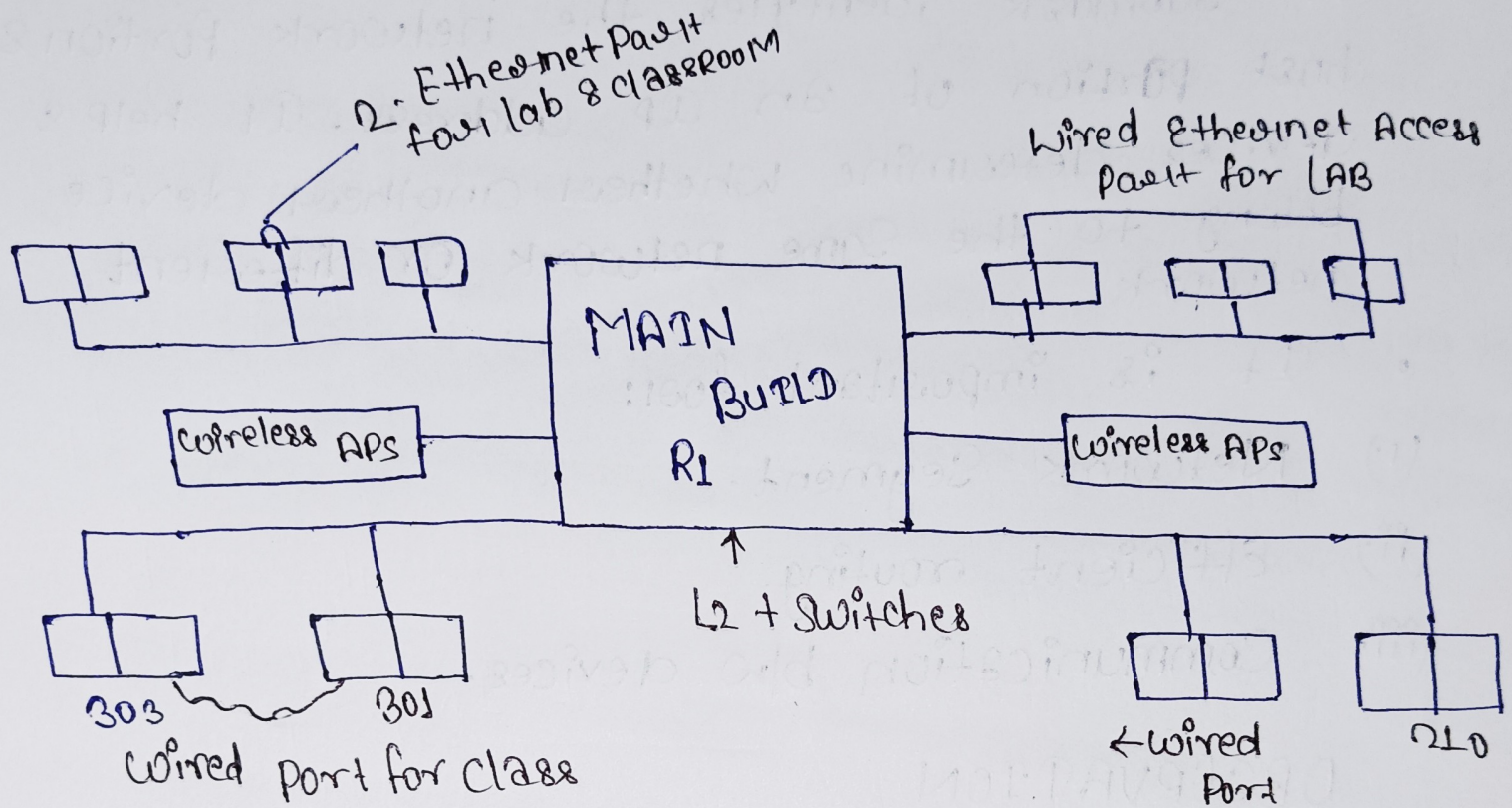
- Every switches act as access layer which helps to provide ~~switch~~ network connectivity to all class room, Lab, faculty Room etc.
- Access layer directly connect:-
 - (i) Ethernet Port.
 - (ii) Wireless AP.
 - (iii) Lab system.
 - (iv) Outdoor wireless AP (ex-01.8)

4) Server Room Architecture / Topology :-



- ↳ $R_1, R_7, R_2, R_3 \rightarrow$ Distribution layer.
- ↳ Server room $\rightarrow$ Core layer
- ↳ $R_1, R_2, R_3, R_4, R_5, R_6, R_7 \rightarrow$ Access layer.

Access Layer Switch (3rd floor - 301)



* Difference b/w Wired & Wireless Speed?

Wired Network

- Fastest Communication.
- Stable Connection.
- Low latency.
- Dedicated Physical Medium.

Wireless Network

- Slowest Communication.
- Affected by distanced Interferences.
- Higher latency.
- Shared radio medium.

↳ Ethernet provides direct physical communication using network cables, resulting in higher speed & stable performance.

C:\Windows\system32\cmd.exe

Subnet Mask : 255.255.255.0
Default Gateway : 10.70.139.52

C:\Users\admin>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

Media State : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Local Area Connection* 1:

Media State : Media disconnected
Connection-specific DNS Suffix . :

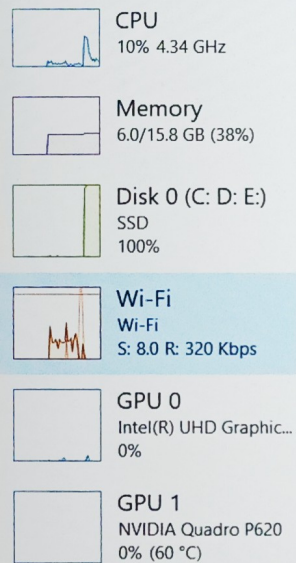
Wireless LAN adapter Local Area Connection* 10:

Media State : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :
Link-local IPv6 Address : fe80::c047:1720:3f7:60e8%3
IPv4 Address. : 10.70.139.104
Subnet Mask : 255.255.255.0
Default Gateway : 10.70.139.52

C:\Users\admin>



Wi-Fi

Throughput



60 seconds

Send
8.0 Kbps

Receive
320 Kbps

Adapter name: Wi-Fi
SSID: realme 9 5G
Connection type: 802.11n
IPv4 address: 10.70.139.104
IPv6 address: fe80::c047:1720:3f7:60e8%3
Signal strength:

Microsoft Windows [Version 10.0.19045.6466]

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C:\Users\admin>ipconfig

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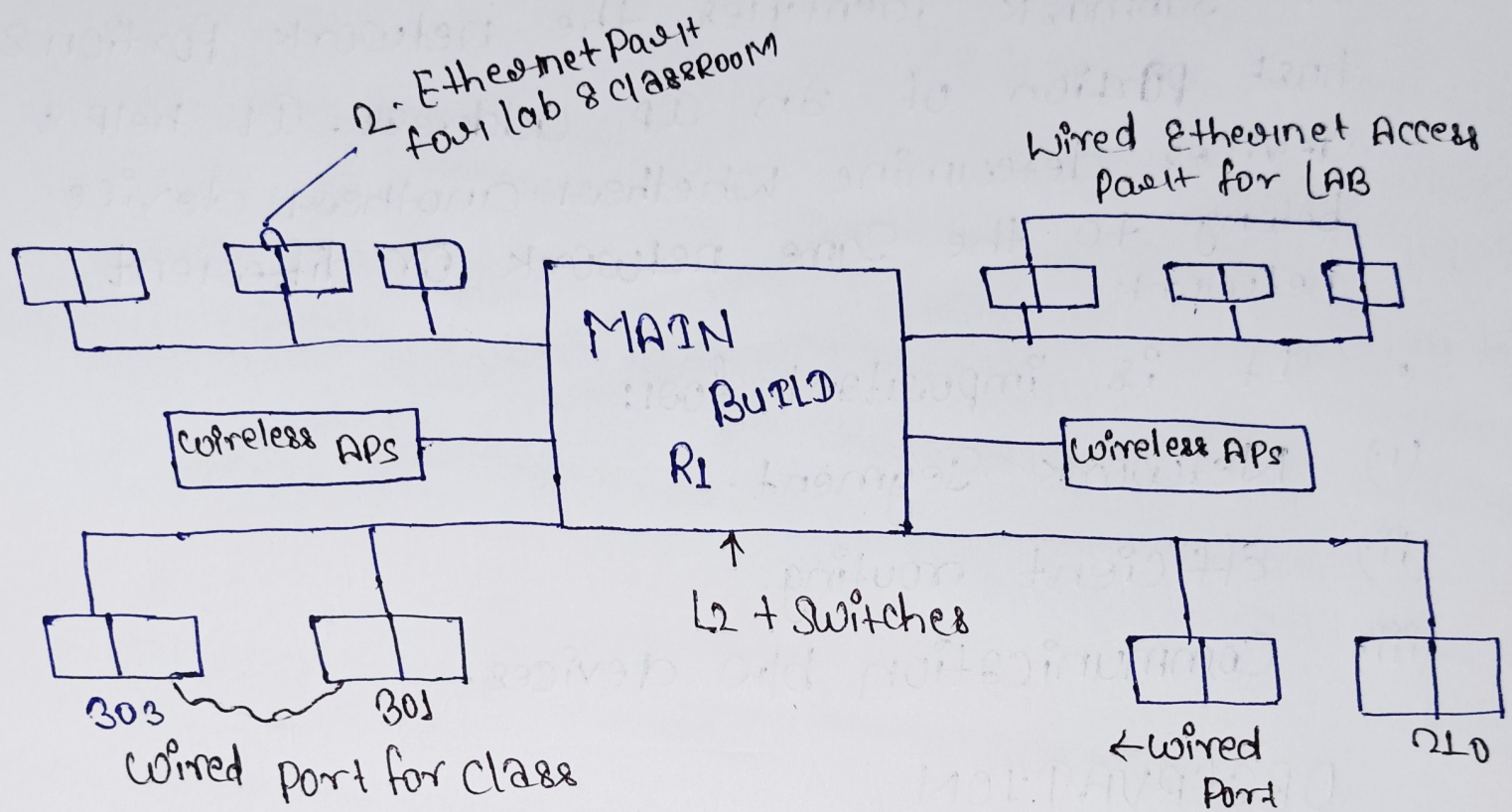
C:\Users\admin>

↳ R₁, R₇, R₂, R₃ → Distribution layer.

↳ Server Room → Core layer

↳ R₁, R₂, R₃, R₄, R₅, R₆, R₇ → Access layer.

Access Layer Switch (3rd floor - 301)



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↳ Ethernet provides direct physical communication using network cables, resulting in higher speed & stable performance.

↳ Wireless Connections use radio waves which are affected by interference, distance, obstacles & multiple connected users.

Q.1) Role of Subnet mask in network?

↳ A Submask identifies the network portion & host portion of an IP address. It helps devices determine whether another device belongs to the same network or different network.

- It is important for:

- (i) Network Segment.

- (ii) Efficient routing.

- (iii) Communication b/w devices.

OBSERVATION

- During the study of the college network infrastructure:-

- (i) A centralized server room was used as network backbone.

- (ii) Two L3 switches were connected in redundant cyclic structure.

- (iii) Managed IP - Link PoE + switches were used in network cabinets.

Advantage of existing Network?

- Scalable architecture.
- Centralized Management.
- Supports PoE + device

- Easy expansion of floors & classroom.
- Support both wireless & wired.
- Redundant link for failover.

Conclusion

↳ The Observed Campus LAN follow a Hybrid hierarchical topology consist of tree, extended star, ring. The network uses manage of PoE switches wireless access point Ethernet port & centralized server infra structure to provide efficient wired & wireless communication, access the campus.